Project – 1

Title - AI-Powered Demand Forecasting and Inventory Optimization System

Submitted By

Team Members:

1. ET23BTIT010 : Zeel Dhorajiya
   gmailid: zeeldhorajiya.it23d1@scet.ac.in ph number: 9924914317

2. ET23BTIT035 : Harnil Navadiya
   gmailid: harnilnavadiya.it23d1@scet.ac.in ph number: 9712746222

3. ET23BTIT037 : Aryan Pandey
   gmailid: aryanpandey.it23d1@scet.ac.in ph number: 8780912463

4. ET23BTIT040 : Dhaval Patel
   gmailid: dhavalpatel.it23d1@scet.ac.in ph number: 93130 82167

5. ET23BTIT049 : Kirtan Savani
   gmailid: kirtansavani.it23d1@scet.ac.in ph number: 9825184800

Under the Guidance of

Prof. (Dr.) Dhruti Sharma

Sarvajanik College of Engineering and Technology

Department of Information Technology

Academic Year: 2026–2027

ABSTRACT

The AI-Powered Demand Forecasting and Inventory Optimization System is a Java Full Stack-based web application designed to help businesses predict future product demand and manage inventory efficiently. Traditional inventory management methods often depend on manual analysis and fixed stock levels, which can result in overstocking, stockouts, increased storage costs, and inefficient resource utilization. The proposed system addresses these challenges by combining Artificial Intelligence and Full Stack development to provide data-driven inventory management.

The system analyzes historical sales records, product information, inventory levels, seasonal patterns, and other relevant factors to forecast future product demand. The forecasting module uses suitable machine learning techniques to identify patterns in historical data and generate demand predictions. Based on these predictions, the system provides inventory recommendations such as optimal stock levels, reorder suggestions, low-stock alerts, and overstock notifications.

The application is developed using Java Full Stack technologies, with Java and Spring Boot used for backend development and RESTful APIs used for communication between the frontend and backend. The frontend provides an interactive dashboard through which administrators or inventory managers can view products, sales information, current inventory, predicted demand, and recommended reorder quantities. MySQL is used as the database for storing product, sales, inventory, user, and forecasting-related information.

The system also provides data visualization through charts and dashboards, allowing users to understand sales trends, demand patterns, and inventory performance more easily. By integrating forecasting and inventory optimization into a single platform, the system reduces dependency on manual calculations and supports faster and more informed decision-making.

The primary objective of the proposed system is to reduce stockouts and excess inventory, minimize inventory holding costs, improve demand prediction accuracy, and optimize inventory levels. The system can be useful for retail stores, e-commerce businesses, wholesalers, and other organizations that manage large quantities of products.

Overall, the AI-Powered Demand Forecasting and Inventory Optimization System demonstrates how Artificial Intelligence can be integrated with Java Full Stack development to create a practical, scalable, and user-friendly business application. The proposed solution helps organizations improve inventory efficiency, respond to changing customer demand, and make more accurate inventory planning decisions.